Forensic Microscopy

Lab 2: Micrometry

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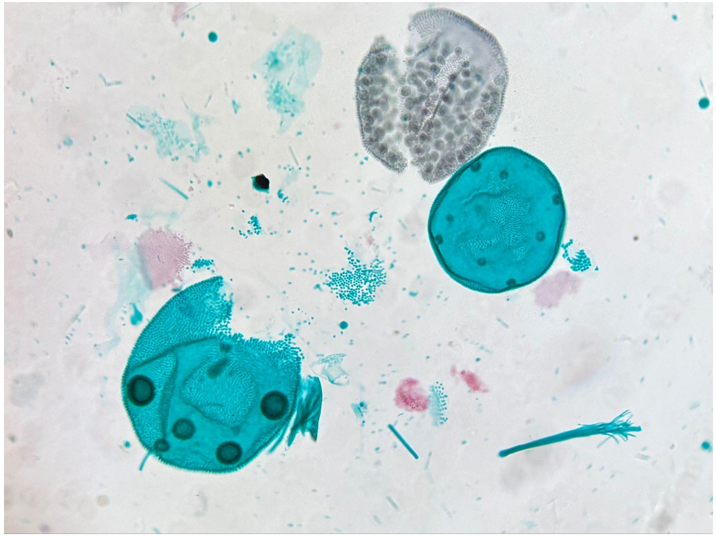
(Grab a Microscope from the black closet)

Where can you find the slides?

- https://aritra-evolves.github.io/FIVS215.html
- Canvas

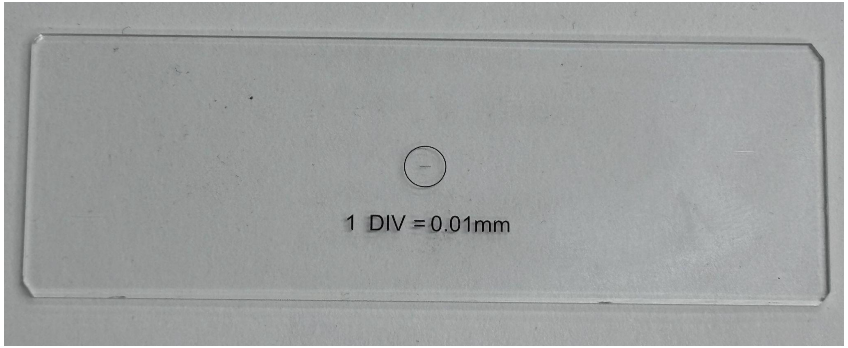
Today’s Agenda

- ❖ Why Micrometry
- ❖ Taking Measurements
- ❖ Calibrating
- ❖ Kohler Review (if wanted)

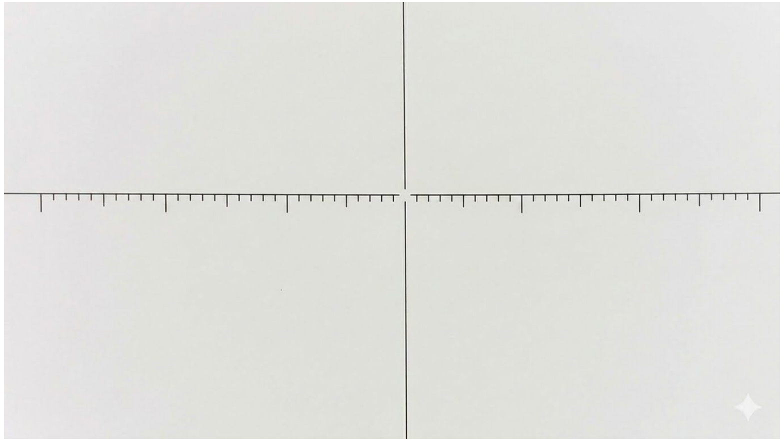


Why micrometry?

- Proper calibration of the microscopes
- Accurate measurement of samples
- Ease in the future



Stage Micrometer



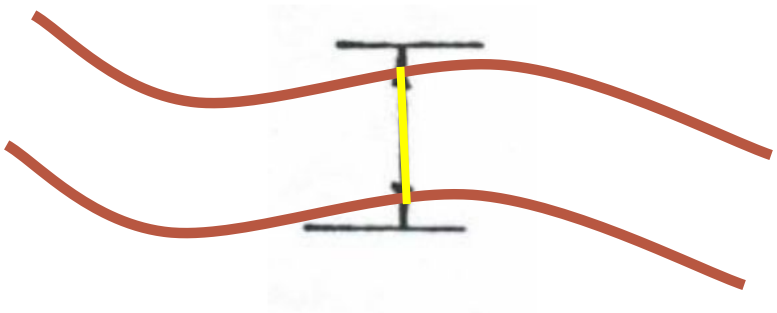
Ocular Micrometer

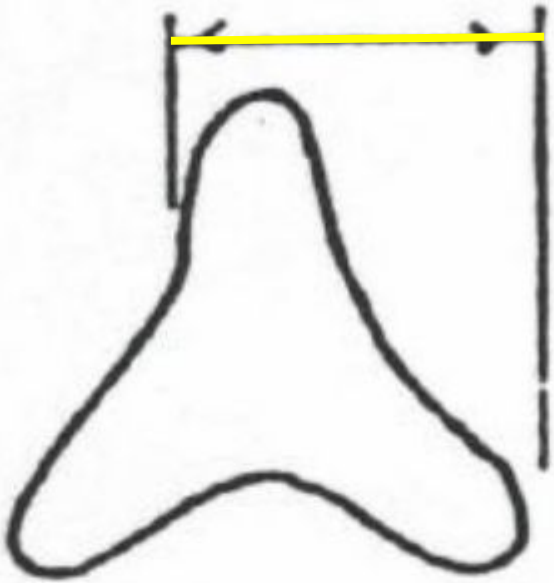
Ocular micrometer picture captured by Aritra B,. retouched with Gemini Nanobanana

Taking Measurements

Two Types of Fibers

Trilobal & Not

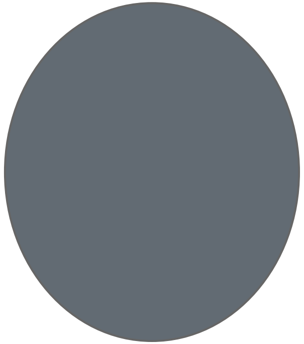




Finding your Dominant Eye

If the object moves, you are looking with your NON-DOMINANT eye

If the object does not move, you are looking with your DOMINANT



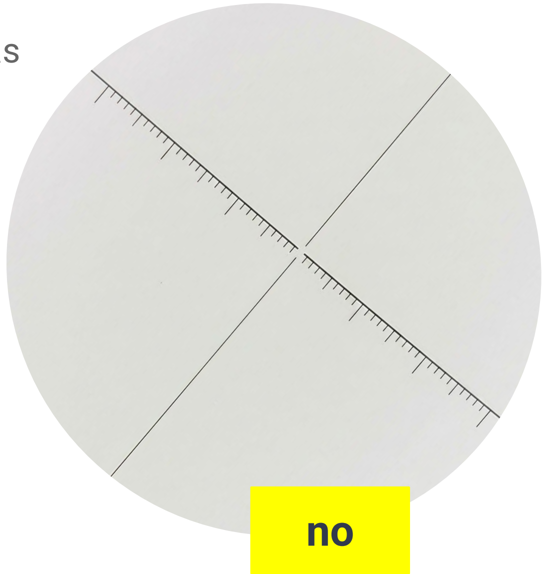
Steps

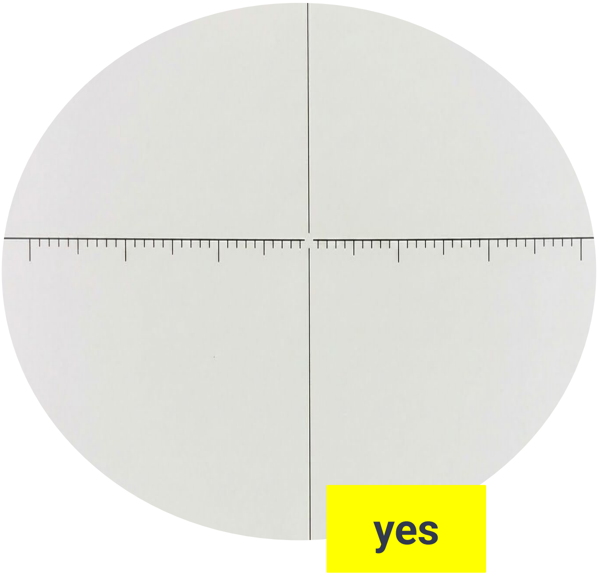
1. Focus on the ocular reticle
    a. Make sure the scale is in your dominant eye, move it if not
2. Swing in the lowest objective
3. Focus on the stage micrometer
4. Align the two micrometers to be parallel
5. Choose two points on the stage micrometer - the farther apart the better
6. Count the number of ocular micrometer marks between the two points
7. Divide using the following equation
8. Repeat with each objective

Step 1 – Focus the Ocular Micrometer

Make sure the ocular micrometer is in clear focus and level

Turn the eyepiece and adjust the diopters as necessary



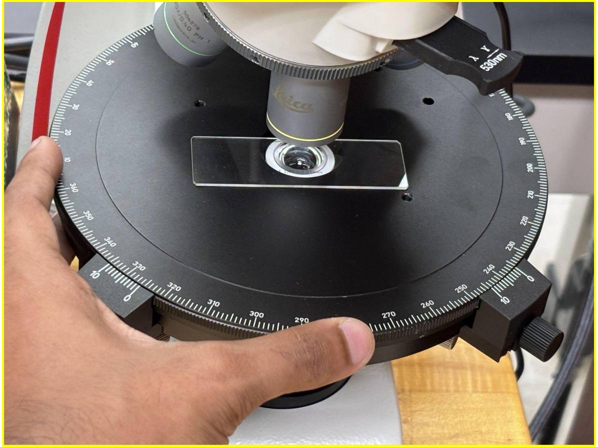


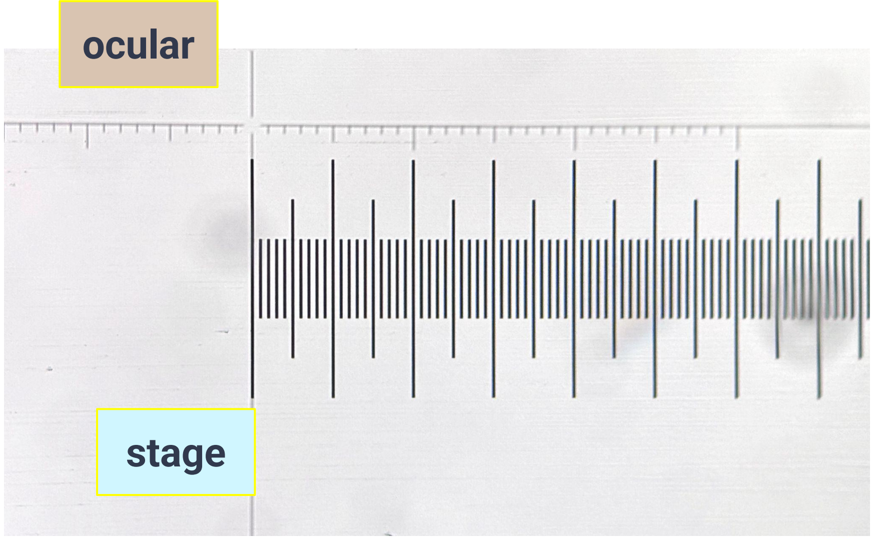
Step 2 - Setting Up the Stage Micrometer

Make sure the lowest objective is swung in (10x)

Place the stage micrometer on the stage and make sure it is in focus

Align the two micrometers so that they are PARALLEL

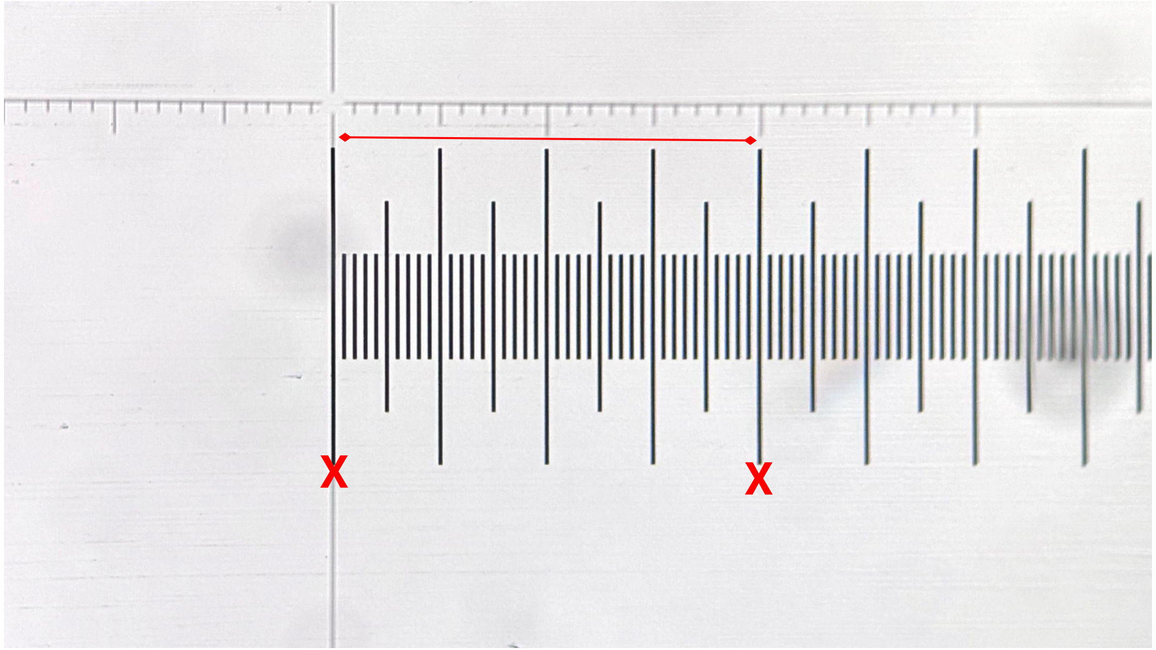




Rotate the stage dial for alignment

Step 3 - Count the Ticks

- Select two points on the STAGE micrometer - the farther apart the better
- Count the number of ticks on the OCULAR micrometer between the two points on the STAGE micrometer
- Write down each count



1 Ocular Micrometer Division= $\frac{\text{\# Stage Micrometer Divisions}}{\text{\# Ocular Micrometer Divisions}}$ x K

K = The Value of the STAGE Micrometer Division in MILLIMETERS

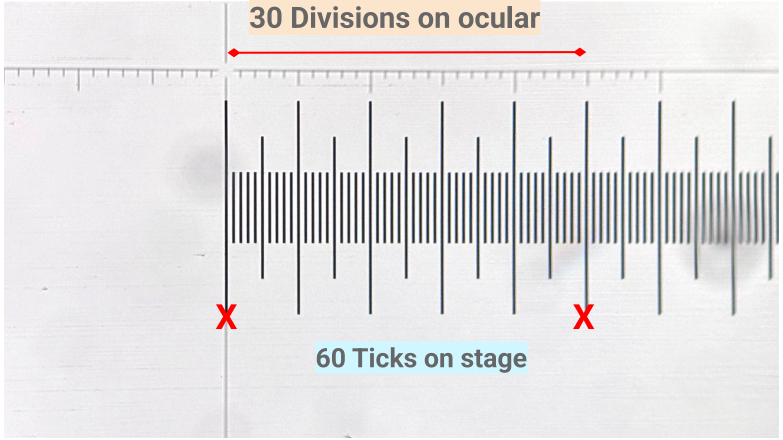
K = 0.01mm (for the sake of this example)

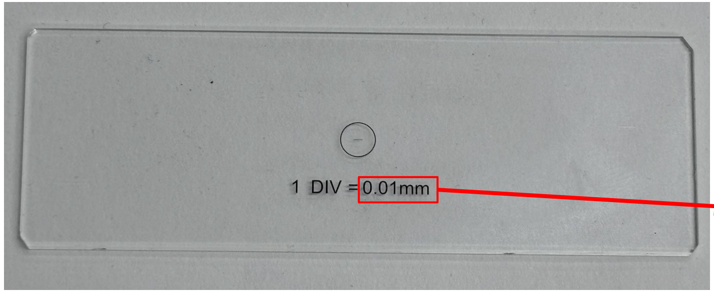
1 OMD = (60/30) x 0.01 mm

= 0.02 mm

1 OMD = 20 micrometer (in this example)

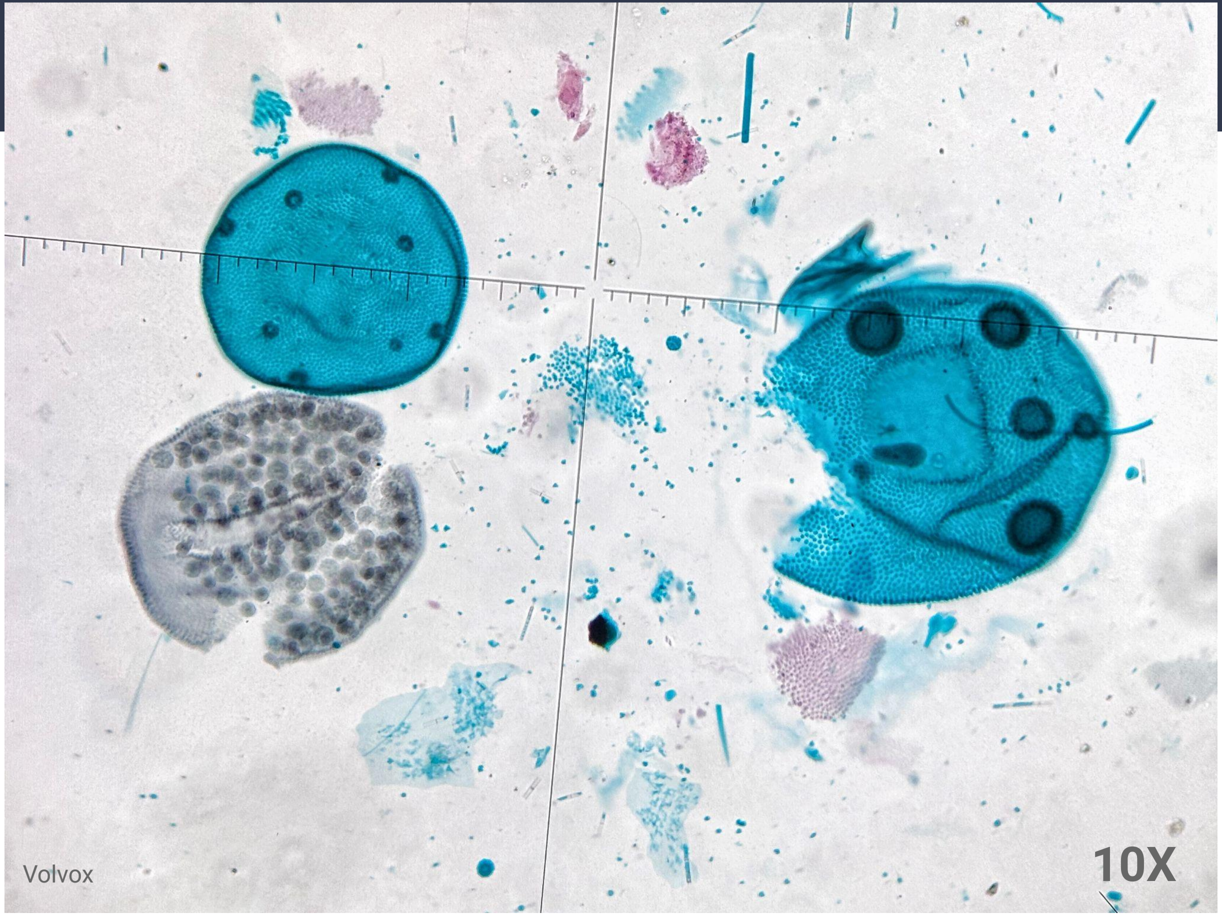
1mm = 1000 micrometer





Value of K

Is it easier to measure now?



Exercise

Set up and calibrate stage and ocular micrometry
For ALL THE OBJECTIBE LENSES (4X, 10X, 40X, 63X).

Write down your calculated values on the index cards and attach it to your micriscope

Keep a copy of the calculated values in your lab notebook.

See you next week!

Stick around if you want to review Kohler again